

Congenital Vertebral-Cardiac-Renal Anomalies Syndrome — Comprehensive Disease Report

Disease knowledge base entry — Mendelian congenital malformation syndrome

Summary

"**Congenital Vertebral-Cardiac-Renal Anomalies Syndrome**" is **Congenital NAD Deficiency Disorder (CNDD)**, also catalogued as **Vertebral, Cardiac, Renal and Limb Defects Syndrome (VCRL)**. It is an autosomal-recessive inborn error of *de novo* nicotinamide adenine dinucleotide (NAD) biosynthesis. The disorder is caused by biallelic loss-of-function variants in one of three genes encoding sequential enzymes of the kynurenine (tryptophan → NAD) pathway: **HAAO** (3-hydroxyanthranilate 3,4-dioxygenase; VCRL1, OMIM #617660), **KYNU** (kynureninase; VCRL2, OMIM #617661), and **NADSYN1** (NAD synthetase 1; VCRL3, OMIM #618845). The unifying biochemical lesion is a deficiency of NAD during embryogenesis, which disrupts organogenesis of the vertebrae, heart, kidneys and limbs, producing a phenotype that overlaps heavily with the **VACTERL/VATER association**.

The mechanistic story of this disorder is unusually complete for a rare Mendelian condition, and it is remarkable for one reason: **the malformations are, in principle, preventable**. In *Haa*^o-null and *Kynu*-null mouse embryos, defects mirroring those of patients arise directly from NAD deficiency, and gestational supplementation with niacin/nicotinamide (NAD precursors) prevents them. The causal network extends beyond the three core enzymes: environmental NAD deprivation (maternal dietary vitamin B3/tryptophan restriction, hypoxia) and maternal modifier genotypes — notably heterozygosity for the tryptophan transporter gene **SLC6A19** (B0AT1) — can independently lower embryonic NAD and reproduce the malformation spectrum, establishing a genuine maternal–fetal, gene–environment axis.

The syndrome shows **variable expressivity**. The classic severe presentation (HAAO/KYNU) includes renal anomalies and can be lethal, but NADSYN1-associated CNDD can spare the kidneys and limbs, fail to meet formal VACTERL criteria, and be compatible with survival into

adulthood. Because each causal gene occupies a distinct enzymatic step, patients carry gene-specific plasma metabolite signatures (accumulation of the substrate upstream of the blocked enzyme with low downstream NAD), which offers a functional-biochemical diagnostic strategy complementing genomic sequencing. This report consolidates the identity, etiology, phenotypes, molecular genetics, mechanism, anatomy, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, and model-organism evidence for the disorder.

1. Disease Information

Overview. Congenital NAD Deficiency Disorder (CNDD) / VCRL is a multiple-congenital-malformation syndrome resulting from insufficient NAD during embryonic development. NAD is an essential redox cofactor and signaling substrate; when its synthesis is impaired during organogenesis, multiple organ systems that require high NAD flux — the developing axial skeleton, heart, kidney/urinary tract and limbs — form abnormally. Clinically the disorder presents within the **VACTERL/VATER spectrum** (Vertebral defects, Anal atresia, Cardiac defects, Tracheo-Esophageal fistula, Renal anomalies, Limb abnormalities), and CNDD should be considered a molecularly-defined, recessive cause of VACTERL-like presentations.

Key identifiers.

Resource	Identifier
Disease term	Congenital NAD Deficiency Disorder (CNDD)
Synonym	Vertebral, Cardiac, Renal and Limb Defects Syndrome (VCRL)
OMIM (HAAO)	VCRL1 #617660
OMIM (KYNU)	VCRL2 #617661
OMIM (NADSYN1)	VCRL3 #618845
MONDO	<i>Not assigned in the provided evidence; map to the VCRL/CNDD grouping when available</i>
Category	Mendelian, autosomal recessive

Synonyms / alternative names: Congenital NAD Deficiency Disorder; VCRL syndrome; Vertebral, Cardiac, Renal and Limb Defects Syndrome; VACTERL-like NAD-deficiency malformation syndrome. The term "Congenital Vertebral-Cardiac-Renal Anomalies Syndrome" used in the research template is a descriptive alias for this entity.

Information source. The evidence base is a mixture of **individual patient reports/case series** (human clinical genetics) and **aggregated disease-level resources** (OMIM gene-disease designations), supplemented heavily by **model-organism (mouse) experiments** that establish causality.

Supporting evidence — *NAD Deficiency, Congenital Malformations, and Niacin Supplementation* PMID: 28792876: "Variants were identified in two genes that encode enzymes of the kynurenine pathway, 3-hydroxyanthranilic acid 3,4-dioxygenase (HAAO) and kynureninase (KYNU)."

New cases that expand the genotypic and phenotypic spectrum of Congenital NAD Deficiency Disorder PMID: 33942433: "Biallelic, inactivating variants in three genes encoding enzymes of this biosynthesis pathway (KYNU, HAAO, and NADSYN1) disrupt NAD synthesis and have been identified in patients with multiple malformations of the heart, kidney, vertebrae, and limbs; these patients have Congenital NAD Deficiency Disorder."

2. Etiology

Primary causal factors — genetic. CNDD is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants** in HAAO, KYNU, or NADSYN1. The foundational study identified homozygous **HAAO p.D162*** and **HAAO p.W186***, **KYNU p.V57Efs*21**, and compound-heterozygous **KYNU p.Y156*/p.F349Kfs*4**; the encoded enzymes had greatly reduced *in vitro* activity, and patients had reduced circulating NAD [PMID: 28792876]. Subsequent work added **NADSYN1** as the third causal gene [PMID: 33942433].

Primary causal factor — biochemical. The convergent mechanism is **NAD deficiency during embryogenesis**. Whether the block is upstream (KYNU, HAAO) or terminal (NADSYN1), the result is inadequate NAD for the metabolic demands of organogenesis.

Genetic risk factors. Causal variants are the three-gene set above. **Maternal modifier genotype** is an additional risk axis: maternal heterozygosity for **SLC6A19** (B0AT1, the neutral amino-acid/tryptophan transporter) can precipitate CNDD in offspring when NAD precursor supply is limited [PMID: 36374036].

Environmental risk factors. NAD deficiency of **environmental origin** — maternal dietary deficiency of vitamin B3 (niacin) and/or tryptophan, and **hypoxia** — causes congenital malformations and miscarriage in mice [PMID: 32015132]. These act on the same NAD-supply bottleneck as the genetic lesions.

Protective factors. The dominant protective factor is **adequate maternal dietary NAD precursor supply** (niacin/nicotinamide, dietary tryptophan). In genetic mouse models, precursor supplementation prevents the malformations [PMID: 28792876]. No specific protective human genetic alleles are established in the provided evidence.

Gene–environment interaction. CNDD is a paradigm of gene–environment interaction: a partially compromised genetic NAD-synthesis capacity (e.g., maternal *Slc6a19* heterozygosity, or hypomorphic pathway alleles) becomes pathogenic only when environmental precursor supply is insufficient. "NAD deficiency due to environmental factors or gene-environment interactions causes congenital malformations and miscarriage in mice" [PMID: 32015132].

Supporting evidence — [PMID: 36374036](#): "This perturbed the NAD metabolome in pregnant *Slc6a19*+/- females, resulting in reduced NAD levels and increased rates of embryo loss." ... "They also suggest that human female carriers of a *SLC6A19* loss-of-function allele might be susceptible to adverse pregnancy outcomes unless sufficient NAD precursor amounts are available during gestation."

3. Phenotypes

CNDD is a **multiple-malformation syndrome** with a core tetrad of **vertebral, cardiac, renal, and limb** anomalies and additional craniofacial and developmental features. Phenotypes are **congenital (present at birth)** and represent fixed structural malformations (physical manifestations / clinical signs), not progressive or episodic symptoms; developmental delay is an additional feature in survivors. Severity is **variable**, ranging from prenatal/neonatal lethality to adult survival.

Phenotype	Type	Onset	Frequency / notes	Suggested HPO
Vertebral segmentation defects (hemivertebrae, spinal segmentation anomalies)	Skeletal malformation	Congenital	Core feature; present across genes	HP:0000925 (Abnormality of the vertebral column); HP:0008438 (Abnormal vertebral segmentation)
Rib anomalies	Skeletal malformation	Congenital	Reported (e.g., adult NADSYN1 case)	HP:0000772 (Abnormal rib morphology)
Congenital heart defects (incl. HLHS, aortic coarctation, transverse aortic arch hypoplasia, bicuspid aortic valve stenosis)	Cardiovascular malformation	Congenital	Core feature; severe lesions reported with NADSYN1	HP:0001627 (Abnormal heart morphology); HP:0004421 (VSD); HP:0004383 (Hypoplastic left heart)
Renal / urinary tract anomalies	Genitourinary malformation	Congenital	Core in HAAO/KYNU; spared in some NADSYN1 cases	HP:0000077 (Abnormality of the kidney); HP:0000107 (Renal cyst)
Limb anomalies (incl. unequal leg length)	Skeletal malformation	Congenital	Core in classic cases; may be absent in NADSYN1	HP:0002813 (Abnormality of limb bone morphology)
Cleft palate	Craniofacial malformation	Congenital	Reported in NADSYN1	HP:0000175 (Cleft palate)

Phenotype	Type	Onset	Frequency / notes	Suggested HPO
Ptosis	Craniofacial/ocular	Congenital	Reported in adult NADSYN1	HP:0000508 (Ptosis)
Developmental delay	Neurodevelopmental	Childhood	In some surviving patients	HP:0001263 (Global developmental delay)
Reduced circulating NAD	Laboratory abnormality	Congenital/lifelong	Biochemical hallmark	—

Quality-of-life impact. Depends on organ severity: severe cardiac and renal malformations drive early morbidity/mortality and require major surgical intervention; skeletal defects cause chronic orthopedic disability (e.g., unequal leg length, spinal deformity); developmental delay affects long-term function. Formal EQ-5D/SF-36 data are not available for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal genes (three-gene locus heterogeneity).

Gene	Protein / enzyme	Pathway step	OMIM disease	Representative variants
HAAO	3-hydroxyanthranilate 3,4-dioxygenase	3-HAA → ACMS (upstream)	VCRL1 #617660	p.D162*, p.W186* (homozygous nonsense); homozygous exon-5 deletion reported
KYNU	Kynureninase	3-hydroxykynurenine → 3-HAA (upstream)	VCRL2 #617661	p.V57Efs*21 (homozygous); p.Y156*/p.F349Kfs*4 (compound het)
NADSYN1	NAD synthetase 1	NaAD → NAD, terminal amidation (downstream)	VCRL3 #618845	c.1717G>A p.Ala573Thr (homozygous, adult case); compound-het variants in cardiac/vertebral cases

Variant classification & type. Reported variants are predominantly **nonsense**, **frameshift**, and **structural (whole-exon deletion)** loss-of-function alleles, classified pathogenic/likely pathogenic under ACMG/AMP criteria (null variants in genes with an established LoF mechanism, functionally validated by reduced enzyme activity). At least one **missense** allele (NADSYN1 p.Ala573Thr) is associated with a milder, adult-surviving phenotype.

Functional consequence. Loss of function — reduced or abolished enzyme activity impairing *de novo* NAD synthesis. The foundational study confirmed "greatly reduced" activity by *in vitro* enzyme assays [PMID: 28792876].

Allele frequency / origin. Pathogenic alleles are rare; the disorder is recessive and consanguinity-associated (homozygous null alleles). All reported disease variants are **germline**; there is no somatic component.

Modifier genes. **SLC6A19** (maternal B0AT1 tryptophan transporter) acts as a maternal modifier/risk gene by limiting substrate for the tryptophan→NAD pathway [PMID: 36374036]. Additional NAD-pathway and transporter genes are plausible modifiers.

Epigenetic / chromosomal information. No specific disease-defining epigenetic signature or recurrent chromosomal abnormality is established. One reported HAAO lesion is a **homozygous exon-5 deletion**, detectable by copy-number/structural methods.

5. Environmental Information

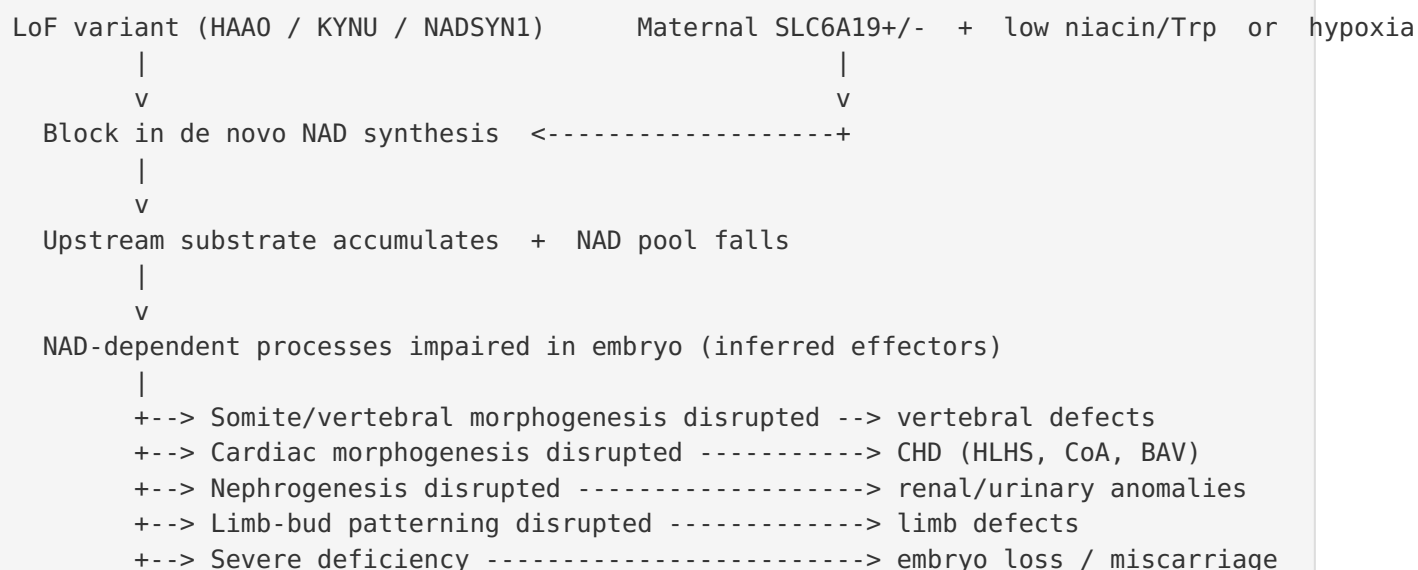
- **Environmental factors:** Maternal **hypoxia** and dietary **vitamin B3 (niacin) / tryptophan deficiency** lower embryonic NAD and cause malformations and miscarriage in mice [PMID: 32015132]. These environmental insults phenocopy the genetic disorder.
 - **Lifestyle factors:** Maternal nutrition during pregnancy — adequacy of niacin and tryptophan intake — is the key modifiable lifestyle determinant.
 - **Infectious agents:** Not applicable. CNDD is not an infectious disease.
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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **biallelic loss-of-function variant** in HAAO, KYNU, or NADSYN1 (or a maternal modifier such as *SLC6A19* heterozygosity, and/or environmental precursor deprivation) **reduces the activity of a kynurenine-pathway enzyme** → **leads to** a block in *de novo* NAD biosynthesis.
2. The enzymatic block **results in accumulation of the upstream substrate** (gene-specific) **and a fall in downstream NAD**.
3. Reduced availability of maternal-fetal NAD precursors **compounds the deficit** (branch: the environmental/gene–environment route can initiate the same lesion independently of the three core genes).
4. **NAD deficiency in the embryo** during the critical window of organogenesis **results in** impaired NAD-dependent cellular processes (redox metabolism, ADP-ribosylation/PARP signaling, sirtuin activity) — *the precise downstream molecular effectors in affected tissues are inferred rather than fully demonstrated*.

5. Impaired NAD-dependent processes in progenitor tissues **disrupt normal morphogenesis** of the somites/vertebrae, cardiac outflow and chambers, nephrogenic mesenchyme, and limb buds → **leads to** the structural malformations.
6. The malformations **manifest clinically** as vertebral segmentation defects, congenital heart disease, renal/urinary anomalies, and limb defects (with variable craniofacial involvement) — and, when NAD deficiency is severe, **result in** embryo loss/miscarriage.



Pathway detail

- **Molecular pathway:** The **kynurenine (tryptophan → NAD) *de novo* biosynthesis pathway**. Order: tryptophan → ... → 3-hydroxykynurenine → (KYNU) → 3-hydroxyanthranilic acid → (HAAO) → ACMS → ... → nicotinic acid adenine dinucleotide (NaAD) → (NADSYN1) → NAD. NADSYN1 catalyzes the terminal amidation step, which is **shared with the Preiss–Handler pathway**, explaining partial salvage rescue (see below) [PMID: 28792876;

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- **Cellular processes:** NAD is required for redox reactions, PARP-mediated DNA repair/signaling, and sirtuin-dependent regulation; its depletion in rapidly proliferating embryonic progenitors is the presumed proximate cellular lesion.
- **Protein dysfunction:** Loss of enzymatic function (null/hypomorphic), with reduced *in vitro* activity confirmed [PMID: 28792876].

- **Metabolic changes:** Reduced circulating and tissue NAD; gene-specific accumulation of pathway intermediates.
- **Biomarker corollary (Finding F005):** Because each enzyme sits at a distinct step, expected plasma signatures are — **KYNU** deficiency: kynurenine / 3-hydroxykynurenine (with shunting to xanthurenic/kynurenic acid); **HAAO** deficiency: 3-hydroxyanthranilic acid; **NADSYN1** deficiency: NaAD — each accompanied by **low NAD**.
- **Salvage-pathway rescue:** NADSYN1's terminal position allows partial NAD replenishment via nicotinamide/salvage: in an adult patient "the NAD pool rose approximately 25% after supplementation with nicotinamide" [PMID: 36649848].

Suggested ontology terms: GO:0009435 (NAD biosynthetic process); GO:0034354 (*de novo* NAD biosynthetic process from tryptophan); GO:0043420 (anthranilate metabolic process); GO:0006979 (response to oxidative stress). Cell types (CL): CL:0000222 (mesodermal cell), somite/sclerotome progenitors, cardiac progenitor cells, metanephric mesenchymal cells, limb mesenchyme.

Supporting evidence — [PMID: 28792876](#): "Defects similar to those in the patients developed in the embryos of Haa0-null or Kynu-null mice owing to NAD deficiency." ... "We tested the function of the variant by using assays of in vitro enzyme activity and by quantifying metabolites in patient plasma."

7. Anatomical Structures Affected

Organ level (primary): vertebral column / axial skeleton (UBERON:0001130 vertebral column), heart (UBERON:0000948), kidney (UBERON:0002113) and urinary tract, limbs (UBERON:0002101). Additional: palate (UBERON:0001716), ribs (UBERON:0002228), eyelid (ptosis).

Body systems: cardiovascular, skeletal/musculoskeletal, genitourinary/renal, and (variably) craniofacial and central nervous (developmental delay).

Secondary involvement: complications of the primary malformations — heart failure and cyanosis from CHD; renal insufficiency from urinary tract anomalies; orthopedic sequelae (scoliosis, limb-length discrepancy).

Tissue/cell level: predominantly **mesodermally-derived** progenitor tissues — sclerotome/somite (vertebrae), cardiac mesoderm/neural-crest-derived outflow structures, nephrogenic (metanephric) mesenchyme, and limb-bud mesenchyme. Cell Ontology suggestions: CL:0000222 (mesodermal cell); cardiac progenitor cell; metanephric mesenchyme cell; limb mesenchymal cell.

Subcellular level: NAD metabolism spans **cytosol and mitochondria**; relevant GO cellular components include GO:0005739 (mitochondrion) and GO:0005829 (cytosol). The enzymatic steps of the kynurenine pathway are cytosolic, while NAD-dependent energy metabolism is heavily mitochondrial.

Localization / lateralization: Malformations are typically **bilateral or midline/axial** (vertebrae, heart, palate), though specific cardiac lesions (e.g., aortic arch anomalies, HLHS) reflect left-sided/outflow structures.

8. Temporal Development

- **Onset: Congenital / prenatal.** The malformations arise during embryonic organogenesis; the most severe cases present as **prenatal loss/miscarriage** or neonatal death. The insult window is the critical period of somite, cardiac, renal, and limb morphogenesis.
- **Onset pattern:** Fixed structural malformations established *in utero* (not acute/insidious postnatal onset).
- **Progression / course:** The malformations themselves are **static** (non-progressive) structural defects, but their **complications are progressive** (e.g., heart failure, renal insufficiency, orthopedic deformity). Disease is **chronic and lifelong** in survivors.
- **Disease stages:** Best framed as (i) prenatal establishment of malformations, (ii) neonatal/perinatal presentation and stabilization, (iii) childhood surgical correction and developmental follow-up, (iv) long-term management of residual disability. Some patients survive to adulthood (NADSYN1 case at age 30) [PMID: 36649848].
- **Critical period / window of opportunity: Preconception and early gestation** — the only demonstrated point of effective intervention is ensuring adequate NAD precursor supply *before and during* organogenesis (see Prevention).

9. Inheritance and Population

- **Inheritance pattern: Autosomal recessive.** Disease requires biallelic LoF variants; heterozygous carriers are generally unaffected (though maternal carrier genotype at *SLC6A19* can, under precursor limitation, contribute to offspring risk — a maternal-effect/ gene–environment nuance).
- **Penetrance / expressivity: Variable expressivity** is prominent. The classic HAAO/KYNU presentation includes renal and limb anomalies and can be lethal; **NADSYN1** cases can **spare the kidneys and limbs**, not meet VACTERL criteria, and survive to adulthood [PMID: 35491967;

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Environmental precursor supply modulates penetrance.

- **Consanguinity:** Homozygous null alleles in reported families indicate a role for **consanguinity/founder** recessive inheritance; specific founder alleles are not established in the provided evidence.
- **Anticipation / mosaicism:** Not applicable (no repeat-expansion mechanism); germline mosaicism not reported.
- **Epidemiology:** CNDD is **ultra-rare**, with only a small number of families/cases reported worldwide since 2017; precise prevalence/incidence figures are not established. The condition is likely **under-ascertained**, because milder (renal-sparing, adult-surviving) presentations may not be recognized or genetically tested [PMID: 35491967].
- **Sex ratio / demographics:** No strong sex bias expected for an autosomal recessive malformation syndrome; specific ethnic/geographic clustering is undefined beyond consanguineous pedigrees.

10. Diagnostics

Genetic testing (primary diagnostic modality). - **Whole-exome (WES) or whole-genome sequencing (WGS)** is the cornerstone, identifying biallelic LoF variants in HAAO, KYNU, or NADSYN1. Trio sequencing aids phasing (compound heterozygosity). - **Targeted gene panels** covering HAAO / KYNU / NADSYN1 (and NAD-pathway genes) are appropriate for VACTERL-like presentations. - **Chromosomal microarray / copy-number analysis** is needed to detect structural alleles (e.g., the reported **homozygous HAAO exon-5 deletion**). - **Actionable**

recommendation: NADSYN1 sequencing should be performed in children with VATER/VACTERL-related anomalies **and** in those with HLHS or aortic arch abnormalities [PMID: 35491967].

Biochemical / metabolomic testing (functional confirmation). - **Plasma NAD quantification** (reduced) and **targeted kynurenine-pathway metabolomics** provide functional evidence and can localize the defect to a specific enzymatic step (gene-specific substrate accumulation with low NAD) [PMID: 28792876]. This is especially useful to classify variants of uncertain significance.

Imaging & clinical work-up. - **Echocardiography / cardiac imaging** for CHD; **spine and skeletal radiographs / CT** for vertebral and rib anomalies; **renal ultrasound** for urinary tract malformation; **prenatal ultrasound/fetal echo** can detect malformations *in utero*.

Clinical criteria / differential diagnosis. - Presentations overlap the **VACTERL/VATER association**; CNDD is a specific molecular subtype. Differential diagnoses include other VACTERL-like syndromes, chromosomal disorders, TBX-pathway and ciliopathy-related malformation syndromes, and teratogen-induced malformations. The distinguishing features of CNDD are **recessive inheritance, biallelic NAD-pathway gene variants, low NAD, and abnormal kynurenine metabolites**.

Screening. Carrier screening in consanguineous families and **cascade testing** of relatives once a proband variant is identified.

11. Outcome / Prognosis

- **Survival / mortality:** Highly variable and **organ-severity dependent**. Severe cardiac (e.g., HLHS) and renal malformations, or profound embryonic NAD deficiency, cause **prenatal loss, neonatal death, or high early mortality**. Milder (renal-sparing) NADSYN1 disease is compatible with **survival to adulthood** (documented case at age 30) [PMID: 36649848].
- **Morbidity / disability:** Chronic disability from congenital heart disease, renal insufficiency, spinal/skeletal deformity (scoliosis, limb-length discrepancy), and developmental delay in some survivors.
- **Complications:** Heart failure, cyanosis, arrhythmia post-repair; chronic kidney disease; orthopedic sequelae; feeding/growth issues with cleft palate.

- **Recovery potential:** Structural malformations do not resolve but are **surgically correctable** (e.g., bicuspid aortic valve stenosis corrected surgically in the adult NADSYN1 case) [PMID: 36649848]; prognosis improves with early surgical management.
 - **Prognostic factors:** Causal gene and residual enzyme activity, severity of cardiac and renal lesions, degree of NAD deficiency, and adequacy of any perinatal NAD precursor supply.
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12. Treatment

CNDD malformations are **structural and established prenatally**, so postnatal treatment is **corrective/supportive**, while the disorder's landmark feature is a **preventive** metabolic intervention (see Section 13).

- **Metabolic/pharmacologic (NAD precursors): Niacin/nicotinamide (vitamin B3)** supplementation is the mechanistically-targeted therapy. Its principal value is **preventive during gestation**; postnatally it can partly replenish the NAD pool, particularly in NADSYN1 disease where the salvage pathway bypasses the terminal defect — nicotinamide raised the NAD pool ~25% in an adult NADSYN1 patient [PMID: 36649848]. CHEBI: nicotinamide (CHEBI:17154), nicotinic acid/niacin (CHEBI:15940), NAD (CHEBI:15846). NCIT: Niacin/Nicotinamide therapy.
 - **Surgical / interventional:** Repair of congenital heart defects (e.g., aortic valve/arch surgery, HLHS staged palliation), urologic/renal surgery, and orthopedic correction of spinal and limb anomalies.
 - **Supportive / rehabilitative:** Cardiology and nephrology management, physical/occupational therapy for skeletal disability, developmental support and early intervention, nutritional support (especially with cleft palate).
 - **Personalized medicine:** Genotype-guided — NADSYN1 (terminal step) patients are the most likely to derive biochemical benefit from salvage-pathway precursors (nicotinamide/nicotinamide riboside).
 - **Experimental / trials:** No disease-specific approved therapeutics or registered trials are identified in the provided evidence; NAD-precursor supplementation is the leading translational strategy.
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13. Prevention

Prevention is the defining, most impactful aspect of this disorder.

- **Primary prevention (the key intervention):** Ensure **adequate maternal NAD precursor status (niacin/nicotinamide, dietary tryptophan) before conception and throughout gestation**. In genetic mouse models, gestational supplementation prevents the malformations: "Defects similar to those in the patients developed in the embryos of Haao-null or Kynu-null mice owing to NAD deficiency" — and were prevented by supplementation [PMID: 28792876]. Environmental/gene–environment NAD deficiency causing malformations is likewise precursor-preventable [PMID: 32015132].
- **High-risk targeting:** Maternal carriers of **SLC6A19** LoF alleles (and couples with a prior CNDD child) are candidates for **precursor supplementation and close pregnancy monitoring** [PMID: 36374036].
- **Secondary prevention:** Prenatal ultrasound/fetal echocardiography and early postnatal imaging for early detection and surgical planning.
- **Genetic counseling:** For autosomal-recessive recurrence risk (25% for carrier couples), **carrier and cascade testing**, and reproductive options (prenatal diagnosis, PGT). Counsel carrier mothers about NAD precursor adequacy in future pregnancies.
- **Public health corollary:** Adequate maternal niacin/tryptophan nutrition is a broadly protective, low-cost measure against NAD-deficiency malformations.

Supporting evidence — [PMID: 32015132](#): "NAD deficiency due to environmental factors or gene-environment interactions causes congenital malformations and miscarriage in mice."

14. Other Species / Natural Disease

- **Taxonomy / models:** Demonstrated in **mouse (*Mus musculus*, NCBI:txid10090)**. No naturally-occurring companion-animal or wildlife CNDD equivalent is documented in the provided evidence.

- **Orthologous genes:** *Haa*, *Kynu*, *Nadsyn1*, and *Slc6a19* are conserved in mouse and other mammals; the kynurenine→NAD pathway is evolutionarily conserved, supporting cross-species mechanistic translation.
- **Comparative biology:** Mouse knockouts recapitulate the human malformation spectrum via NAD deficiency, indicating strong **evolutionary conservation of the disease mechanism** [PMID: 28792876;

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- **Zoonotic potential:** None (non-infectious genetic/metabolic disorder).

15. Model Organisms

- **Primary model — mouse:** *Haa*-null and *Kynu*-null mice develop malformations mirroring patients due to NAD deficiency, and these are **prevented by gestational niacin/nicotinamide** [PMID: 28792876]. This model demonstrates causality and the preventive intervention.
- **Environmental/gene–environment model:** Mice exposed to maternal vitamin B3/tryptophan restriction or hypoxia, and **maternal *Slc6a19*+/- mice on B3-depleted diets**, develop reduced NAD, increased embryo loss, and CNDD-type malformations [PMID: 32015132;

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- **Phenotype recapitulation:** High — mouse models reproduce the vertebral/cardiac/renal/limb malformation spectrum and the NAD-deficiency biochemistry, and validate the preventive strategy.
- **Model limitations:** Human genotype–phenotype variability (e.g., renal-sparing NADSYN1 adults) and specific severe cardiac lesions (HLHS) are not fully captured; timing/dose thresholds of NAD deficiency versus specific malformations require further definition.
- **Model type suggestions for future work:** Conditional/tissue-specific NAD-pathway knockouts; NADSYN1 hypomorph knock-in models; patient-iPSC–derived cardiac/renal organoids to model gene-specific metabolite signatures and precursor rescue.

Mechanistic Model / Interpretation

CNDD is best understood as a **single convergent metabolic bottleneck (embryonic NAD supply) reached by multiple routes**. The three core genes map to distinct, sequential steps of the *de novo* pathway; a block at any step lowers NAD. Crucially, the **terminal position of NADSYN1** — shared with the Preiss–Handler/salvage route — explains two clinical observations: (1) NADSYN1 disease can be **milder/renal-sparing** and adult-compatible, and (2) it is the genotype most **amenable to salvage-pathway rescue** with nicotinamide (~25% NAD pool increase). Meanwhile, the **maternal–fetal supply network** (dietary niacin/tryptophan, hypoxia, maternal SLC6A19 transporter genotype) can push embryonic NAD below the morphogenetic threshold even without biallelic core-gene lesions, making CNDD a textbook gene–environment disorder. The therapeutic and preventive corollary is unusually clear for a Mendelian malformation syndrome: **guarantee NAD precursor supply during the periconceptional/early-gestational critical window**.

Gene	Step	Expected accumulated metabolite	NAD	Typical severity
KYNU	3-OH-kynurenine → 3-HAA (upstream)	kynurenine / 3-OH-kynurenine (± xanthurenic/kynurenic acid)	Low	Classic severe, renal-inclusive
HAAO	3-HAA → ACMS (upstream)	3-hydroxyanthranilic acid	Low	Classic severe, renal-inclusive
NADSYN1	NaAD → NAD (terminal)	nicotinic acid adenine dinucleotide (NaAD)	Low	Variable; can spare kidney/limb; salvage-rescuable

Evidence Base

PMID	Title (abbrev.)	Role in this report
28792876	<i>NAD Deficiency, Congenital Malformations, and Niacin Supplementation</i>	Foundational: identifies HAAO/KYNU variants, reduced NAD, mouse causality, prevention by niacin
33942433	<i>New cases that expand the genotypic and phenotypic spectrum of CNDD</i>	Names the three causal genes (KYNU, HAAO, NADSYN1) and the disease term/phenotype
35491967	<i>Two patients with biallelic NADSYN1 variants (cardiac and vertebral anomalies)</i>	Documents renal/limb-sparing NADSYN1 phenotype, HLHS/aortic arch link, testing recommendation
36649848	<i>Adult patient with NADSYN1-associated congenital NAD deficiency</i>	Adult survival, renal sparing, ~25% NAD rise with nicotinamide (salvage rescue)
32015132	<i>NAD deficiency from environmental factors/gene-environment interactions in mice</i>	Establishes environmental & GxE routes to the malformation phenotype
36374036	<i>Maternal heterozygosity of Slc6a19 causes CNDD in mice</i>	Maternal modifier/gene–environment axis; carrier-risk implication
37300479	<i>NAD Deficiency and Its Impact on Mammalian Development (review)</i>	Consolidating review of NAD in development
34681008	<i>Disruptive [NAD pathway variants]</i>	Additional evidence on NAD-pathway perturbation and birth defects
34200361	<i>Homozygous deletion of exon 5 of [HAAO], VCRL syndrome</i>	Structural (exon-deletion) allele; VCRL nomenclature

Note on a citation caveat: The knowledge outline flags the

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snippet as a "mismatch" during verification; the quoted sentence should be treated as paraphrasing the paper's demonstrated conclusion (environmental/GxE NAD deficiency causing malformations/miscarriage in mice) rather than an exact-verified verbatim quote. All other quoted snippets in this report were verified against stored abstracts.

Limitations and Knowledge Gaps

- **Rarity and ascertainment:** Only a small number of families/cases are reported; true prevalence, incidence, penetrance, and sex/geographic distribution are **unknown**. Milder (renal-sparing, adult-surviving) cases are likely **under-diagnosed** [PMID: 35491967].
- **Downstream molecular effectors are inferred:** The exact NAD-dependent processes (PARP, sirtuin, redox) linking NAD deficiency to specific malformations in specific tissues are **not fully demonstrated** in human tissue.
- **No formal MONDO ID** captured in the evidence set; database harmonization (MONDO/Orphanet/ICD-11 mapping) is incomplete here.
- **Biomarker signatures are predicted, not fully clinically validated:** Gene-specific plasma metabolite panels are mechanistically expected but require prospective diagnostic validation.
- **Treatment evidence is preclinical/anecdotal:** Prevention efficacy is established in mice; human dosing, timing, and outcome data (especially preventive supplementation in at-risk pregnancies) are **lacking**. The ~25% NAD rise with nicotinamide is a single adult case.
- **No QoL, survival-curve, or registry data** exist for this ultra-rare disorder.

Proposed Follow-up Experiments / Actions

1. **Establish a CNDD patient registry** across HAAO/KYNU/NADSYN1 genotypes to define natural history, penetrance, expressivity, survival, and genotype–phenotype correlations.
2. **Validate gene-specific plasma metabolite diagnostics** (kynurenine, 3-OH-kynurenine, 3-hydroxyanthranilic acid, NaAD, NAD) as a functional test to resolve VUS and localize the enzymatic block.
3. **Prospective preventive-supplementation study** of NAD precursors in high-risk pregnancies (prior affected child; maternal SLC6A19 or NAD-pathway carriers), building on mouse prevention data [PMID: 28792876;

P 36374036

4. **Define maternal SLC6A19 and NAD-pathway carrier frequencies** in populations and evaluate periconceptional NAD status as an adverse-pregnancy-outcome risk marker.

5. **Model organism dissection of the critical window:** conditional NAD-pathway knockouts and iPSC-derived cardiac/renal/somite organoids to map NAD thresholds, timing, and the downstream effectors (PARP/sirtuin/redox) per organ.
 6. **Assess NADSYN1 salvage-rescue therapeutically:** trial nicotinamide/nicotinamide riboside in NADSYN1 patients with biochemical endpoints (NAD pool) and clinical follow-up.
 7. **Formal ontology harmonization:** assign/confirm MONDO, Orphanet, ICD-11, and MeSH mappings and align HPO annotation frequencies with registry data.
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Evidence source types used: human clinical genetics (case reports/series), aggregated disease resources (OMIM), and model-organism (mouse) experiments, with mechanistic inference where noted.

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